

5G RRC Signaling Storm Mitigation

via a Proactive Digital Twin Architecture
with Continuous Calibration



Omer Dikyol

Department of Computer Engineering
Istanbul Technical University



Outline

01

Problem Definition & Research Gap

Why availability matters; the storm; control-loop latency

02

System Model

MMPP/M/c/K queue and gNB RACH parameters

03

Proposed Method

Calibrated Digital Twin with EMA + asymmetric step rule

04

Results & Evaluation

Mirroring error, queue stability, delay CDF, ϵ -sensitivity

05

Conclusion

Findings, O-RAN deployment, future work

Why 5G availability matters

5G NR underpins mission-critical services — every device starts with an RRC connection request through the gNB RACH.

<10 ms

Round-trip latency required by autonomous vehicles

5×9

Reliability target for remote surgery and industrial IoT

100s–1000s

UEs sharing a single gNB simultaneously

gNB RACH at a glance

c = 54

preamble channels

K = 200

buffer capacity

Q ≥ K

requests dropped immediately



The RACH is a shared, limited resource with no built-in overload anticipation. RACH failure ⇒ no data plane, all services blocked.

What is an RRC signaling storm?

RRC and the RACH

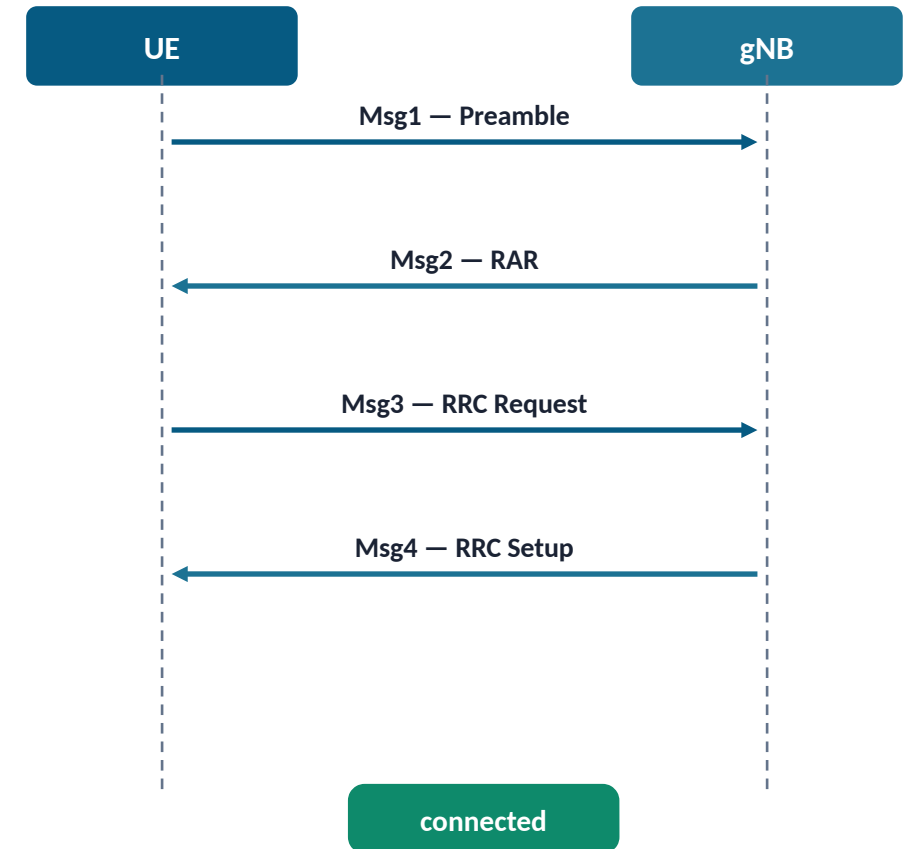
RRC is the 3GPP Layer-3 protocol that manages connection setup. No data flows without a completed RRC connection.

How a storm forms

- Botnet floods Msg1 simultaneously, overwhelming $c = 54$ preamble slots
- Preamble collision $\Rightarrow$ both UEs fail $\Rightarrow$ retry
- gNB buffers pending Msg3 (RRC Requests) up to $K = 200$
- At $\lambda_{\text{high}} = 500$ req/s, buffer fills in 0.4 s
- Dropped devices retry — load never subsides

Net effect: RACH becomes a denial-of-service point — legitimate UEs cannot complete connection setup.

RACH procedure (Msg1–Msg4)



The RRC signaling storm attack

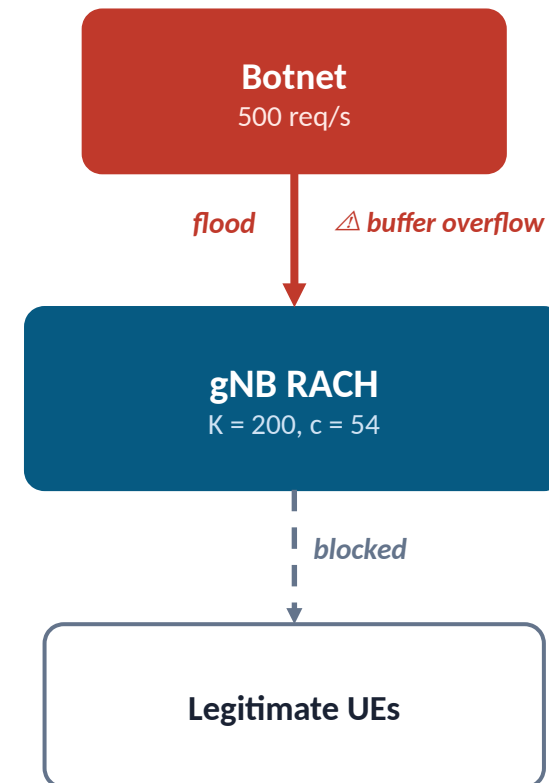
Attack mechanism

- Adversary coordinates a botnet of compromised devices
- Each device sends RRC connection requests at high rate
- Combined rate: $\lambda_{\text{high}} = 500 \text{ req/s}$ vs. normal $\lambda_{\text{low}} = 10 \text{ req/s}$ (50×)
- RACH buffer saturates within seconds
- Access requests are dropped — QoS failure

Termed “RRC Signaling Storm” in the literature [b1].

Impact: All connected services lose access until the attack subsides.

Attack flow



PROBLEM

Control-loop latency in ACB/UAC

Current defence: Access Class Barring (ACB) — 3GPP TS 38.331 UAC framework.



Fundamental causality constraint

ACB throttles only after observing $Q_{\text{real}} > 0.8K$. During the build-up, requests are silently dropped.

Empirical evidence

Kotuliak et al. (2021): RACH exceeded delay budget under high load in commercial 5G NR deployments [b6].

Why existing approaches fall short



Reactive ACB / UAC

Strength

Simple, standards-compliant

Limitation

Reacts after saturation; inherent control-loop latency



ML / DNN predictors

Strength

High accuracy on stationary traffic

Limitation

$O(W)$ inference, $W \sim 10^5 - 10^7$; ms latency — impractical on gNB hardware



Standard Digital Twin

Strength

Proactive control possible

Limitation

Model drift under non-stationary traffic; mirroring error grows unbounded



Gap: no DT approach suppresses model drift via an online calibration loop during the attack interval. This work closes that gap.

MMPP/M/c/K queue

Two-state CTMC: Markov-Modulated Poisson Process



Generator matrix

$$Q = \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix} ; \quad \text{sojourn} \sim \text{Exp}(|Q_{ii}|)$$

Exact CTMC dynamics in DES — no matrix exponential needed.

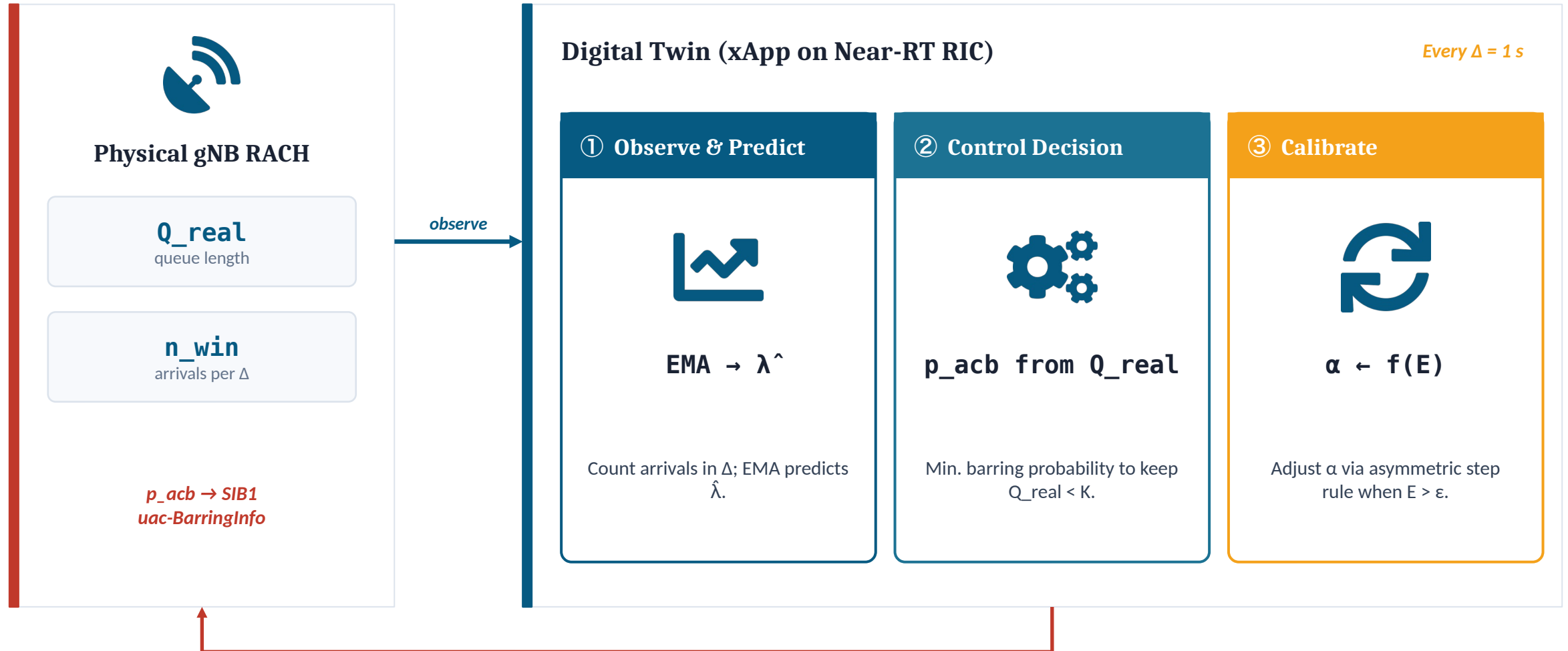
Physical gNB RACH

Parameter	Value
Preamble channels	$c = 54$
Buffer capacity	$K = 200$
Service time	$\text{Exp}(1)$



Stationary blocking governed by classical M/c/K formulas;
non-stationarity comes from MMPP modulation.

Calibrated Digital Twin architecture



EMA predictor and ACB control policy



EMA arrival-rate predictor

$$\text{EMA}_t = \alpha \cdot \lambda_{\text{cur}} + (1 - \alpha) \cdot \text{EMA}_{\{t-1\}}$$

$$\lambda_{\text{cur}} = n_{\text{win}} / \Delta$$

- Initial weight $\alpha_0 = 0.20$
- α is dynamic — updated by the calibration loop (Stage 3)

Key insight: p_{acb} rises before Q_{real} reaches K — proactive throttling, not reactive.



Stochastic barring probability

$$p_{\text{acb}} = \max(0, \min(1, 1 - (K - Q_{\text{real}}) / \max(\text{EMA}_t \cdot \Delta, 1e-9)))$$

- Minimum probability to prevent $Q_{\text{real}} \geq K$ next step
- Applied as a Bernoulli trial per request
- Maps to uac-BarringInfo in SIB1 (3GPP TS 38.331)

Continuous calibration loop

State-mirroring error

$$E(t) = | Q_{\text{real}}(t) - Q_{\text{virtual}}(t) |$$

Asymmetric step rule

```
if E > ε :  
    α ← min(α_max, α + δ+)  
else:  
    α ← max(α_min, α - δ-)
```

ε**10**

calibration threshold

δ⁺**0.05**

fast attack response

δ⁻**0.01**

slow decay (stability)

Attack onset: δ⁺ = 0.05 (fast)

- EMA becomes high-frequency
- Outdated observations discarded quickly

Normal traffic: δ⁻ = 0.01 (slow)

- Statistical stability preserved
- Avoids oscillation around ε

Calibrated DT control loop

 Every Δ control step

```

1.  $\lambda_{\text{cur}} \leftarrow n_{\text{win}} / \Delta$ 
2.  $\text{EMA}_t \leftarrow \alpha \cdot \lambda_{\text{cur}} + (1 - \alpha) \cdot \text{EMA}_{\{t-1\}}$ 
3.  $d \leftarrow \max(\text{EMA}_t \cdot \Delta, 1e-9)$ 
4.  $p_{\text{acb}} \leftarrow \max(0, \min(1, 1 - (K - Q_{\text{real}}) / d))$ 
5.  $Q_{\text{virtual}} \leftarrow \text{clip}(Q_{\text{virtual}} + \text{EMA}_t \cdot (1 - p_{\text{acb}}) \cdot \Delta - c \cdot \Delta, 0, K)$ 
6.  $E \leftarrow |Q_{\text{real}} - Q_{\text{virtual}}|$ 
7. if  $E > \varepsilon$  :  $\alpha \leftarrow \min(\alpha_{\text{max}}, \alpha + \delta^+)$ 
8. else      :  $\alpha \leftarrow \max(\alpha_{\text{min}}, \alpha - \delta^-)$ 
9.  $\text{drop\_rate} \leftarrow p_{\text{acb}}$ 

```

Complexity

Calibrated DT

$O(1)$

Fixed arithmetic · 4 state variables · μs control budget

DNN predictors

$O(W)$

$W \sim 10^5 - 10^7$ · ms latency · impractical on gNB

State: $\{ \text{EMA}_t, Q_{\text{virtual}}, \alpha, n_{\text{win}} \}$

Simulation setup



Experimental configuration

- Discrete-event simulation: SimPy, Python 3.10
- $T_{sim} = 1000$ s per run
- $N = 10$ independent seeds (Monte Carlo)
- 95 % CI: $\bar{x} \pm 1.96 \cdot \hat{\sigma} / \sqrt{N}$

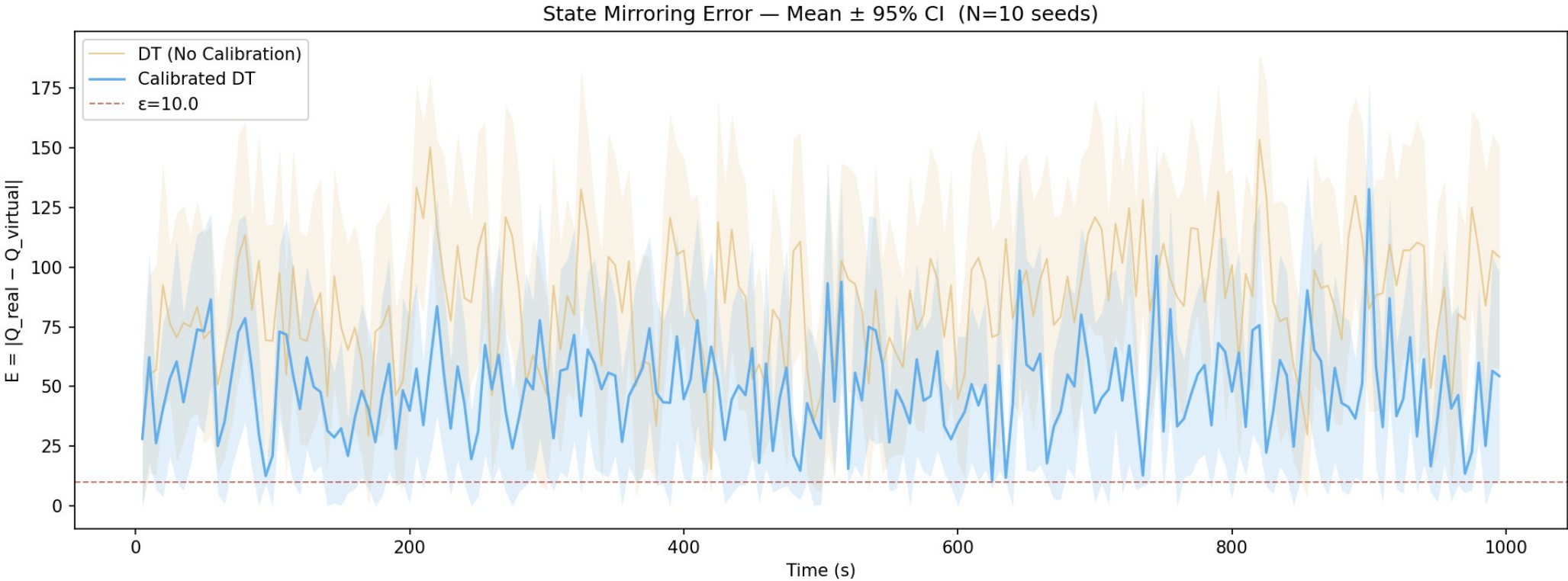
Three methods compared

- Reactive Baseline** *threshold 0.8 K, drop 0.5*
- DT no-calibration** *α fixed*
- Calibrated DT** *asymmetric step rule*

Key parameters

Parameter	Symbol	Value
Preamble channels	c	54
Buffer capacity	K	200
Normal arrival rate	λ_{low}	10 req/s
Burst arrival rate	λ_{high}	500 req/s
Control interval	Δ	1 s
Calibration threshold	ϵ	10
Initial EMA weight	α_0	0.20

State-mirroring error — 42% reduction



42 %

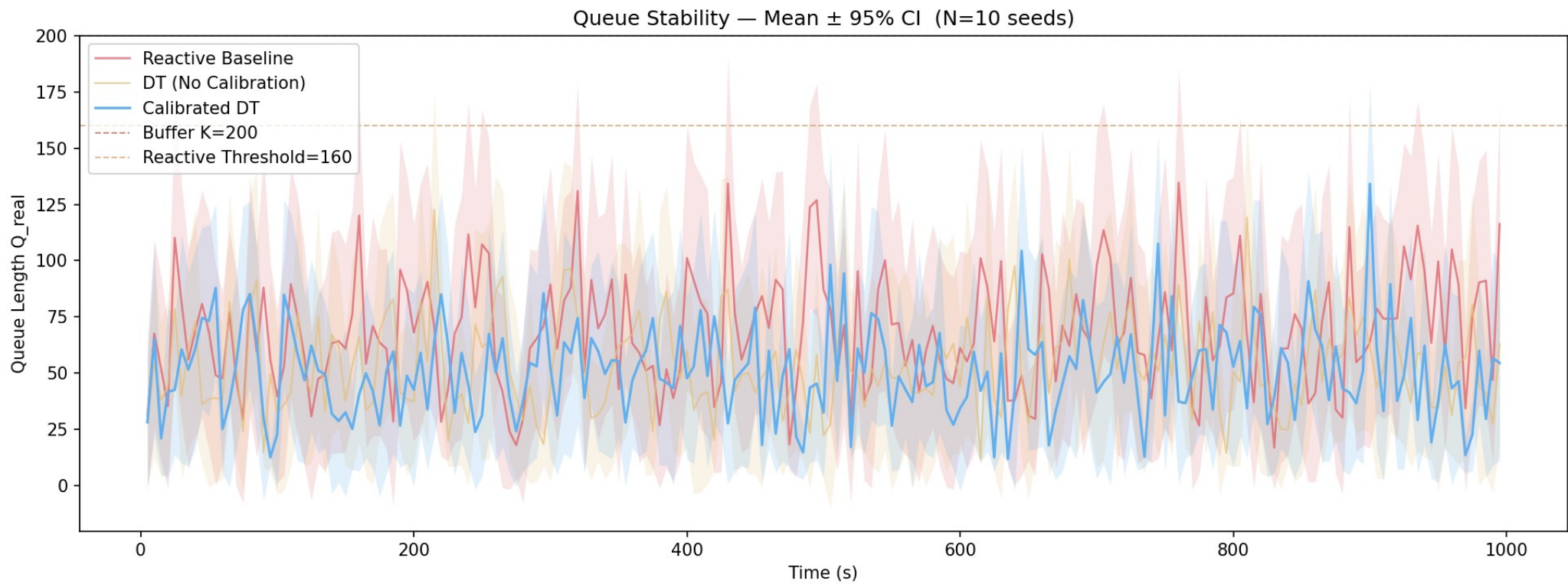
reduction in E_{mean}

E_mean (N = 10, 95 % CI)	
DT (no calib.)	85.03 $\pm$ 4.40
Calibrated DT	49.14 $\pm$ 3.09

Takeaway

Calibration suppresses model drift continuously during the burst — CIs do not overlap.

Queue stability



P_{success} (N = 10, 95 % CI)

Reactive	0.310 ± 0.015
DT (no calib.)	0.301 ± 0.011
Calibrated DT	0.309 ± 0.019

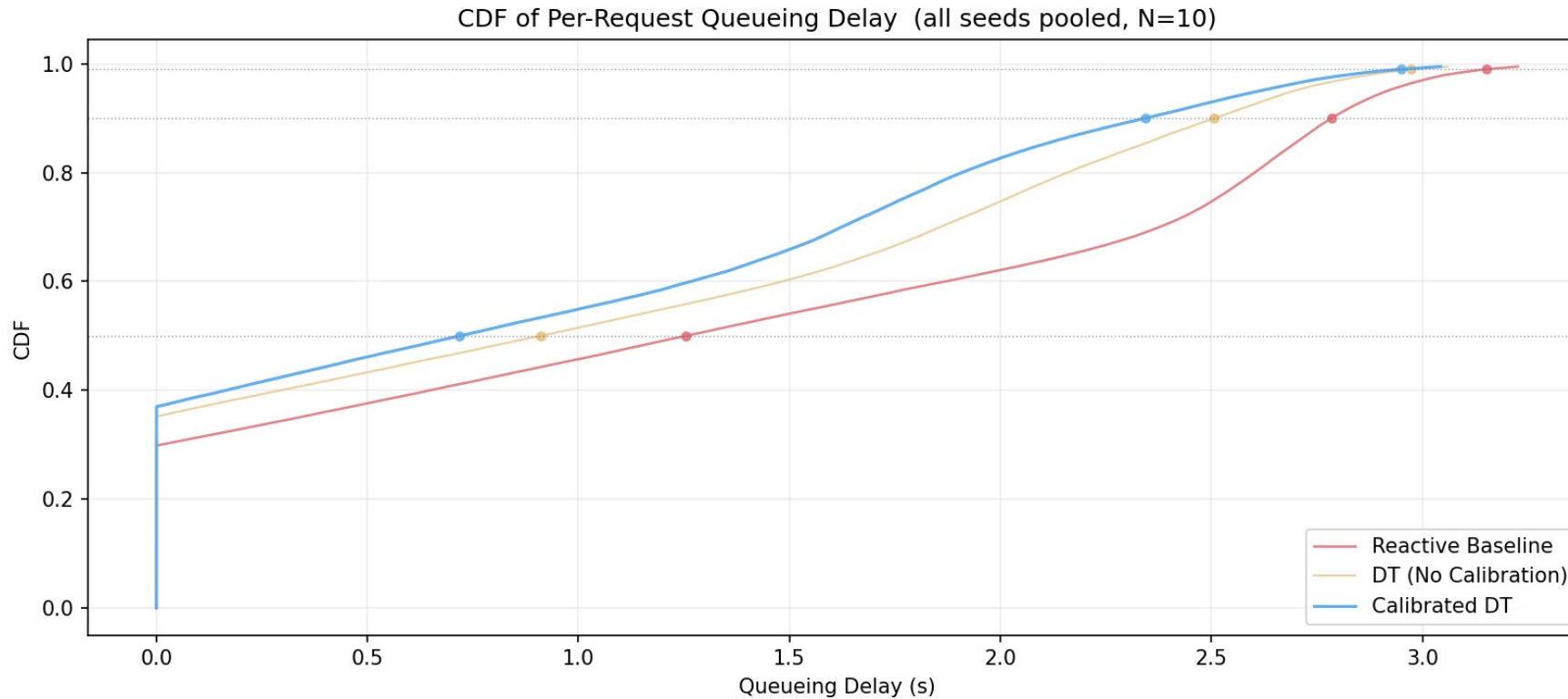


statistically equivalent to baseline; no throughput penalty

Interpretation

- Reactive only acts after Q > 0.8 K
- Calibrated DT raises p_{acb} before saturation

Queueing delay — CDF



CDF left-shift

Calibrated DT curve sits left of the others — lower delay across all reported percentiles.

Tail suppressed

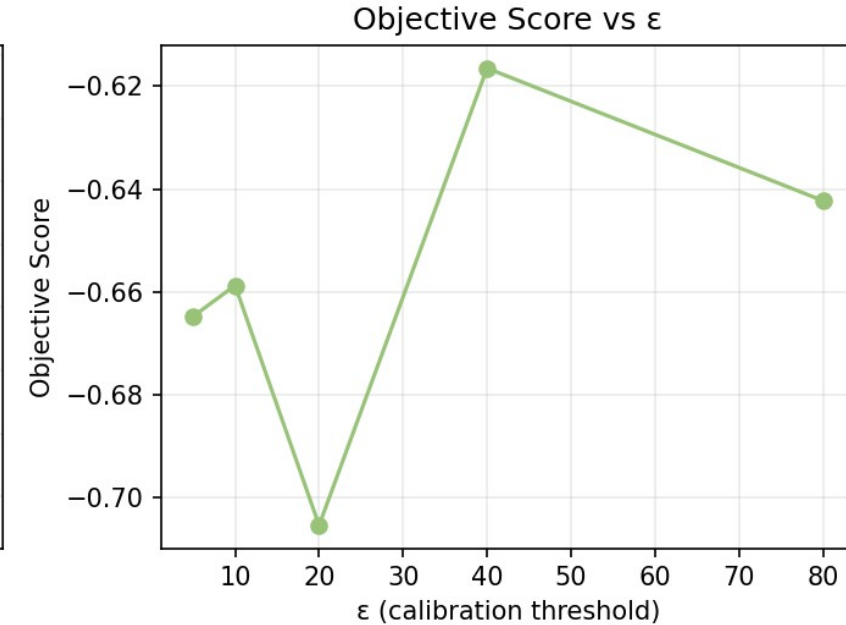
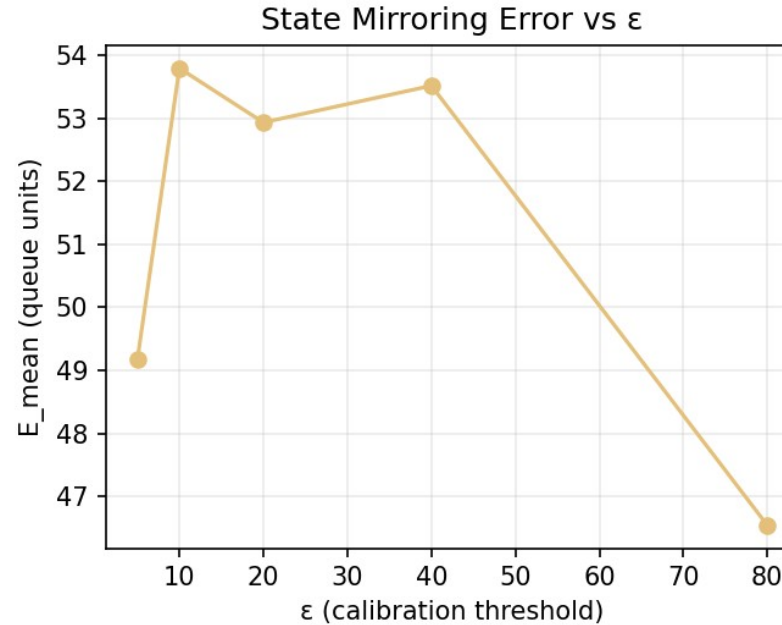
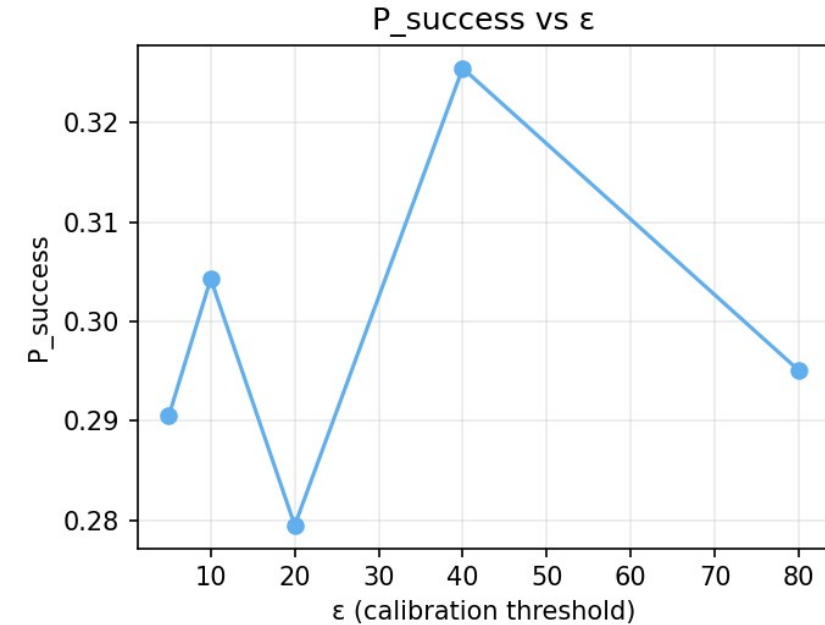
P90 / P99 queueing delay roughly halved — critical for URLLC latency budgets.

Mechanism

- Reactive admits requests into a deeply loaded queue
- Calibrated DT keeps Q_{real} low → short waits

ϵ -sensitivity analysis

Epsilon Sensitivity Analysis (Calibrated DT, seed=42)



Tested values

$\epsilon \in \{ 5, 10, 20, 40, 80 \}$

Single-seed sweep (seed = 42)

Interpretation

- $\epsilon = 5 \rightarrow$ over-calibrates on normal traffic
- $\epsilon = 80 \rightarrow$ calibration rarely triggers
- $\epsilon = 40 \rightarrow$ best composite in this sweep

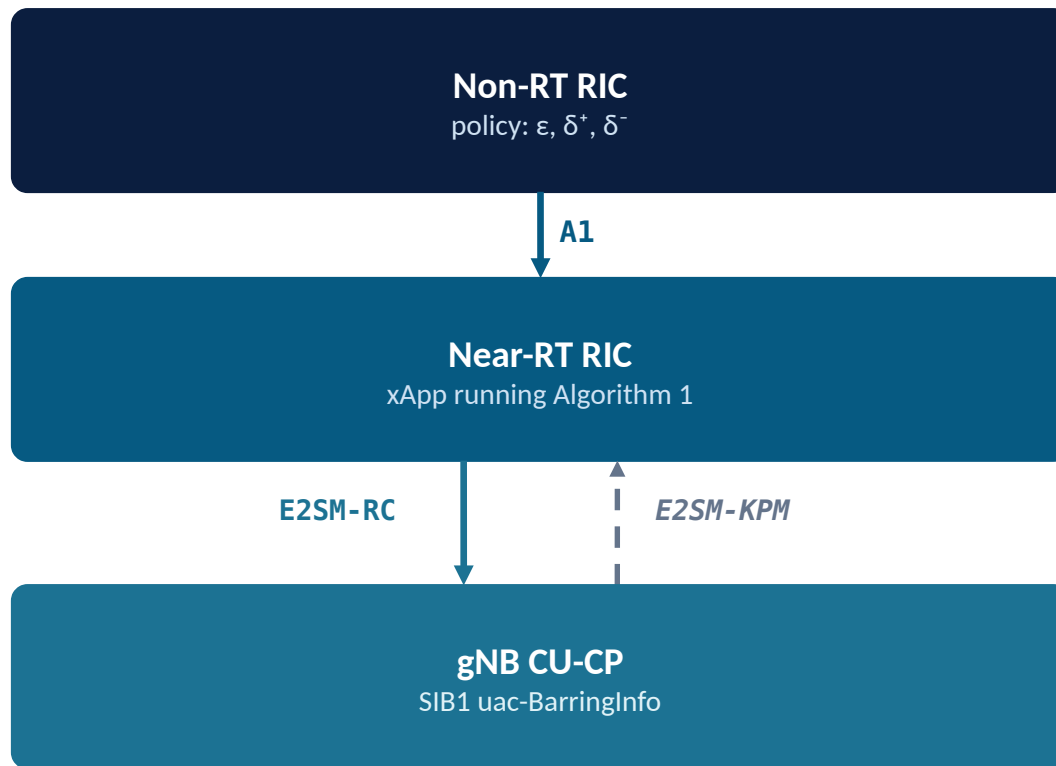
Adopted default

$\epsilon = 10$

conservative — just above the normal-traffic noise floor

$O(1)$ complexity and O-RAN integration

O-RAN integration



Deployment notes

- $\Delta = 1$ s fits Near-RT RIC range (10 ms – 1 s)
- No protocol modifications required
- Vendor-independent xApp — standard E2 service models
- $O(1)$ per step — fixed-cost arithmetic, suitable for tight gNB control budgets after hardware benchmarking

Calibrated DT runs in μ s vs. ms-class DNN inference — practical for commercial gNB software.

Findings and future work

42 %

reduction in state-mirroring error vs. uncalibrated DT

$\approx$

P_success statistically equivalent to baseline; P90/P99 delay reduced

$O(1)$

deployable in commercial gNB software via O-RAN xApp



Proactive control in non-stationary 5G environments requires both a sound control policy and tight virtual-physical alignment.

Multi-cell extension

Current model treats one gNB cell independently. Real deployments involve handover correlations and inter-cell interference — a distributed DT with inter-cell state sharing is required.

RL-based calibration

Fixed δ^+ and δ^- are vulnerable to slow ramp-up attacks. An RL agent can learn step sizes dynamically using state-mirroring error E as a reward signal.



Questions?

5G RRC Signaling Storm Mitigation via a Proactive Digital Twin Architecture with Continuous Calibration

Omer Dik Yol

dik yol25@itu.edu.tr

BLG 632E · ITU

Selected references

- [b1]** D. Rupprecht et al., "Breaking LTE on Layer Two," Proc. IEEE S&P, May 2019, pp. 1121–1136.
- [b2]** B. H. Lee et al., "Enhanced Random Access for Massive-MTC," IEEE IoT J., vol. 8, no. 8, 2021.
- [b3]** M. Grieves and J. Vickers, "Digital Twin: Mitigating Unpredictable Emergent Behavior," Springer, 2017.
- [b4]** 3GPP, "NR; RRC Protocol Specification," TS 38.331, Rel. 17.3.0, 2022.
- [b5]** L. Miuccio et al., "Joint Control of Random Access for Massive MTC in 5G NR," IEEE IoT J., vol. 7, no. 6, 2020.
- [b6]** M. Kotuliak et al., "Performance Evaluation of 5G NR Non-Standalone Network," IEEE Access, vol. 9, 2021.
- [b7]** S. K. Sharma and X. Wang, "Toward Massive MTC in Ultra-Dense Cellular IoT Networks," IEEE COMST, 2020.
- [b8]** R. Dong et al., "Deep Learning for Radio Resource Allocation in 5G," IEEE TWC, vol. 20, no. 4, 2021.
- [b9]** C. Zhang et al., "Deep Learning in Mobile and Wireless Networking: A Survey," IEEE COMST, 2019.
- [b10]** H. X. Nguyen et al., "Digital Twin for 5G and Beyond," IEEE Commun. Mag., vol. 59, no. 2, 2021.
- [b14]** M. Polese et al., "Understanding O-RAN," IEEE COMST, vol. 25, no. 2, 2023.